

Juan Pablo Alfonso

ASTROPHYSICS GRADUATE STUDENT · ASTROSTATISTICS · GALAXY FORMATION & EVOLUTION

Astronomical Institute, Tohoku University

6-3 Aramaki, Aoba Ward, Sendai, Miyagi, Japan 980-8578

✉ juanpabloalfonso@astr.tohoku.ac.jp | 🏠 juanpabloalfonso.github.io/ | 📧 [juanpabloalfonso](#) | 🌐 [juan-pablo-alfonso-7b9623201](#)

Summary

My research lies in the area of **galaxy evolution and formation** within astrophysics. My research in particular is very programming/computationally heavy. I like describing my area of research as an **intersection between astronomy, statistics, and computer science**, specifically **machine learning and deep learning**. My research has a heavy emphasis on machine learning and **artificial neural networks** to help answer research questions associated with galaxy evolution and formation. Currently most of my research is focused on trying to understand how galaxy structures are formed, and how they change with time.

Education

Tohoku University (東北大学)

PHD IN ASTRONOMY

Sendai, Japan

Oct. 2022 - Oct. 2027

- Supervisors: Prof. Masayuki Akiyama & Dr. Kartheik Iyer (Columbia University/Flatiron Institute)

University of Toronto

HONS. BSC. IN ASTRONOMY AND PHYSICS

Toronto, Canada

Sept. 2017 - Jun. 2021

- Math minor and Philosophy minor
- Thesis Supervisor: Dr. Mubdi Rahman

Research Experience

Tohoku University (東北大学)

GRADUATE STUDENT RESEARCHER

Sendai, Japan

Oct 2022- Present

- Working on applying **machine learning and deep learning techniques** to problems in various areas of astronomy
- Primarily interested in extra-galactic astronomy
- Supervisor: Prof. Masayuki Akiyama & Dr. Kartheik Iyer (Columbia University/Flatiron Institute)

Columbia University

VISITING RESEARCHER

New York, USA

Feb 2025 - April 2025

- Working with Dr. Kartheik Iyer to understand the morphology of galaxies in the Illustris-TNG50 simulation suite

Center for Computational Astrophysics (The Flatiron Institute)

VISITING RESEARCHER

New York, USA

Feb 2025 - April 2025

- Working with Dr. Francisco Villaescusa-Navarro to understand how dark matter models and cosmological models in the DREAMS project effect galaxy morphology

University of Toronto

VISITING RESEARCHER

Toronto, Canada

Nov 2024 - Feb 2025

- Working with Dr. Mike Walmsley and Prof. Josh Speagle to study the morphology of galaxies targeted by the Euclid space telescope

University of Toronto

SURP RESEARCHER/ RESEARCH SCHOLAR

Toronto, Canada

May 2021 - Sept 2022

- Working on **training a convolutional neural network (CNN) to predict the star forming history of the galaxies in the SDSS-IV MaNGA survey from their visual image and photometric information**, while studying what features of the galaxies the neural network is focusing on to make the predictions with the use of **CAM methods**.
- Project Supervisor: Dr. Kathreik Iyer (Dunlap Institute)

University of Toronto

UNDERGRADUATE RESEARCHER

Toronto, Canada

Aug. 2020 - Apr. 2021

- Investigated morphological changes in galaxies as they evolve in time, by studying their internal kinematics using the MaNGA survey.
- Analysis being done using Python based code with **heavy use of machine learning algorithms such as DB scan and PCA** from the **sci-kit learn** library. Full write up of project can be found [here](#)
- Project Supervisor: Dr. Mubdi Rahman (Dunlap Institute)

First-Author Publications

Selected Presentations and Talks

12th Galaxy Evolution Workshop

CONTRIBUTED TALK

*Seoul, S. Korea**Aug. 24-28, 2026*

Beyond Hubble

CONTRIBUTED TALK

*Cambridge, UK**Apr. 20-24, 2026*

SUPER-IRNET Workshop 2026

CONTRIBUTED TALK

*Osaka, Japan**March 16-19, 2026*

11th Galaxy Evolution Workshop

CONTRIBUTED TALK

*Nagoya, Japan**Aug. 25-29, 2025*

Cosmology and Galaxy Astrophysics with Simulations and Machine Learning 2024

POSTER PRESENTATION

*New York, USA**Dec. 9-12, 2024*

10th Galaxy Evolution Workshop

CONTRIBUTED TALK

*Taipei, Taiwan**Aug. 6-9, 2024*

SUPER-IRNET Workshop 2024

CONTRIBUTED TALK

*Beppu, Japan**Jul. 23-26, 2024*

IPMU Astrophysics Lunch Talk

INVITED TALK

*Kashiwa, Japan**Jun. 18, 2024*

ELT Science in Light of JWST

POSTER PRESENTATION

*Sendai, Japan**Jun. 3 - Jun. 7, 2024*

Recipes to Regulate Star Formation at All Scales

POSTER PRESENTATION

*Baltimore, USA**Apr 15-19, 2024*

East Asian Young Astronomers Meeting 2024

POSTER PRESENTATION

*Chiang Mai, Thailand**Jan 30 - Feb 2, 2024*

AI-Driven Discovery in Physics and Astrophysics

FLASH TALK PRESENTER

*Kashiwa, Japan**Jan 22-26, 2024*

Machine Learning in Astronomical Surveys (IAP/CCA 2023)

POSTER PRESENTATION (REMOTE)

*Paris, France & New York, USA**Nov 27 - Dec 1, 2023*

Resolving Universe

POSTER PRESENTATION

*Tokyo, Japan**Nov 5-10, 2023*

Statistical Challenges in Modern Astronomy VIII

POSTER PRESENTATION

*State College, USA**June 12-16, 2023*

ART (Astrostats) Meeting

SEMINAR TALK

*Toronto, Canada**Mar. 30, 2023*

9th Galaxy Evolution Workshop

CONTRIBUTED TALK

Kyoto, Japan

Feb. 21, 2023

HSC-AGN Meeting 2022

CONTRIBUTED TALK

Kagoshima, Japan

Dec. 2, 2022

SDSS Collaboration Meeting 2021

PRESENTER IN DATA 1 SERIES AND LIGHTING TALK 1 SERIES

Baltimore, USA

Aug. 11, 2021

Awards

SUPER-IRNET Travel Funding

POSTER PRESENTER

Baltimore, USA

Apr 15 - Apr 19, 2024

- Secured full travel and accommodation funding to present a poster at a star-formation conference at STScI. (¥550,000)

Graduate Program on Physics and the Universe (GPPU) Fellowship

GPPU STUDENT

Sendai, Japan

April 2024 - Oct 2027

- Fellowship for the graduate program at Tohoku University for Physics and Astronomy PhD students. The fellowship provides full financial funding to students, and includes travel funding. (≈ ¥11,500,000)

SUPER-IRNET Travel Funding

POSTER PRESENTER

Tokyo, Japan

Nov 5 - Nov 10, 2023

- Secured full travel and accommodation funding to present a poster at an ALMA and JWST conference at Waseda University. (¥76,940)

SUPER-IRNET Travel Funding

POSTER PRESENTER

State College, USA

June 4 - June 17, 2023

- Secured full travel and accommodation funding to attend an astrostats summer school and to do a poster presentation at an astrostats conference at Penn State. (¥734,525)

2022 Embassy Recommended Research MEXT Scholar

GRADUATE STUDENT

Sendai, Japan

Oct 2022 - Oct 2024

- The Japan Ministry of Education, Culture, Sports, Science and Technology (in Japanese: Monbukagakusho, English acronym: MEXT) offers scholarships to Canadian students wishing to pursue their studies at a Japanese university. For graduate studies (non-degree, masters or doctorate) in any university subjects. Scholarship provides: full tuition coverage, monthly allowance, round-trip between Canada and Japan, and Japanese language training. Information can be found [here](#) (≈ ¥10,000,000)
- Offered extension to PhD course, but declined due to a fellowship being awarded

2021 SURP Top Poster Award

POSTER PRESENTER

Toronto, Canada

Aug 13, 2021

- Won top poster award for poster conference that was held to summarize all research projects carried out in the SURP program at the University of Toronto in the summer of 2021 (\$50)